

Ferhat Elmas

Distributed systems and storage engineer

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Updated
September 2026

PROFILE

Systems engineer with 15+ years of experience building distributed data platforms, storage systems, high-throughput APIs, and developer infrastructure. Delivered zero-downtime migrations and multi-million-RPS rollouts, and grew a globally distributed engineering team.

CORE EXPERTISE

Systems Distributed systems, storage, caching, consistency, sharding, zero-downtime migrations

Languages Go, Rust, Python, TypeScript/JavaScript

Data and infrastructure PostgreSQL, object storage, RocksDB, CockroachDB, DynamoDB, Kafka, Redis/Valkey, Memcached, Consul, Kubernetes, Elasticsearch, CDNs

EXPERIENCE

Supabase

Amsterdam, Netherlands

Storage Engineer

February 2026 - Present

- Build and operate [Supabase Storage](#), an S3-compatible object storage product backed by multi-tenant PostgreSQL. Use customer workloads and production telemetry to set product and architecture priorities.
- Led a large database and runtime refactor. Replaced Knex with pg, introduced transport-neutral contracts, and moved Storage to a multi-worker architecture while preserving REST, S3, and TUS behavior.
- Cut CPU, allocations, and connection churn across S3 routing, SigV4 uploads, authorization, logging, metrics, TLS, and tenant pools, improving throughput and lowering compute and database cost per request.
- Expanded the product with vector storage across pgvector and S3 Vectors, adding OrioleDB support, pagination, provider-specific limits, and cross-backend acceptance coverage.
- Built foundations for object versioning and lifecycle policies. Used progressive, migration-aware rollouts across existing tenants and hardened TUS and database-connection failure paths.

Cloudflare

Amsterdam, Netherlands

Senior Systems Engineer

February 2023 - January 2026

- Designed KV's tiered-cache and multi-worker architecture to abstract multiple storage engines while balancing P95 latency, consistency, bandwidth, and replica cost.
- Moved KV to a hybrid storage architecture combining a custom engine with R2 through a live data migration, using gradual rollout and rollback at multi-million RPS.
- Moved KV's REST API from core services to the edge, improving performance and rate limiting while adding batch operations.
- Built a KV sync engine on sharded Cloudflare Queues to make global synchronization across multiple storage systems faster and more reliable.
- Replaced full rescans for KV billing with incremental accounting and added edge-cached account tags for R2 billing.
- Built a strongly consistent global cache for R2 metadata in Go with Memcached and Consul, preserving S3 compatibility and sub-millisecond latency requirements.

EXPERIENCE CONTINUED

Stream Inc.

Principal Engineer

Backend Tech Lead

Senior Engineer

Amsterdam, Netherlands

June 2022 - January 2023

April 2021 - May 2022

November 2019 - March 2021

- Migrated data systems from Keevo, PostgreSQL, DynamoDB, and Elasticsearch to CockroachDB without downtime while schemas and product features continued to change.
- Built chat campaigns that delivered and tracked millions of messages, with adaptive throttling, analytics, and actionable error reporting.
- Shipped link enrichment, a custom S3 and CloudFront CDN, and push-notification infrastructure across multiple providers and tenants.
- Developed search, feed ranking, real-time updates, automatic sharding, and billing pipelines under high write volume and tight latency constraints.

AgFlow

CTO

Senior Developer

Geneva, Switzerland

December 2016 - October 2019

February 2015 - November 2016

- Advanced from Senior Developer to CTO and grew the globally distributed engineering team from 2 to 11.
- Built automated ingestion for email, HTML, Excel, PDF, and OCR sources, combining machine learning, synthetic data, and human review to handle incomplete and ambiguous inputs.
- Created a data API across JSON, RPC, Excel, FIX, and SQL while protecting the confidentiality of source data.
- Introduced Kubernetes, Rook, TimescaleDB, Pachyderm, Drone, and Prometheus; migrated the web client from CoffeeScript to TypeScript and launched the first Flutter clients.

EARLIER EXPERIENCE

Unit9

Backend and Infrastructure Developer

Built the Real Foodie backend and processed 3.5 TB of Twitter data for CES brand analytics.

London, United Kingdom

August 2014 - January 2015

Indico at CERN

Software Developer

Modernized Indico, extracted services, added polyglot storage, and helped move it to Python 3.

Geneva, Switzerland

March 2013 - August 2014

Community Software Lab

Software Developer

Refactored social-service search engines, built a database-agnostic ORM, and normalized schemas.

Massachusetts, United States

June 2011 - February 2012

Bogazici University

Software Developer and Teaching Assistant

Built traffic-simulation tools and taught systems programming.

Istanbul, Turkey

February 2011 - June 2011

Commencis

Software Developer

Built monitoring, digital-signage, mobile-distribution, banking-log, and XML tooling.

Istanbul, Turkey

January 2009 - June 2011

EDUCATION

Master of Science, Computer Science

Swiss Federal Institute of Technology Lausanne, EPFL

Internet Computing specialization. Thesis: *New Storage System for Indico, Event Management at CERN*.

Projects included Nokia Mobile Data Challenge and *Squall*, an online query-processing engine for Twitter Storm.

2014

Lausanne, Switzerland

Bachelor of Engineering, Computer Engineering

Bogazici University

Thesis: *Social Analysis on DeviantART*.

2011

Istanbul, Turkey